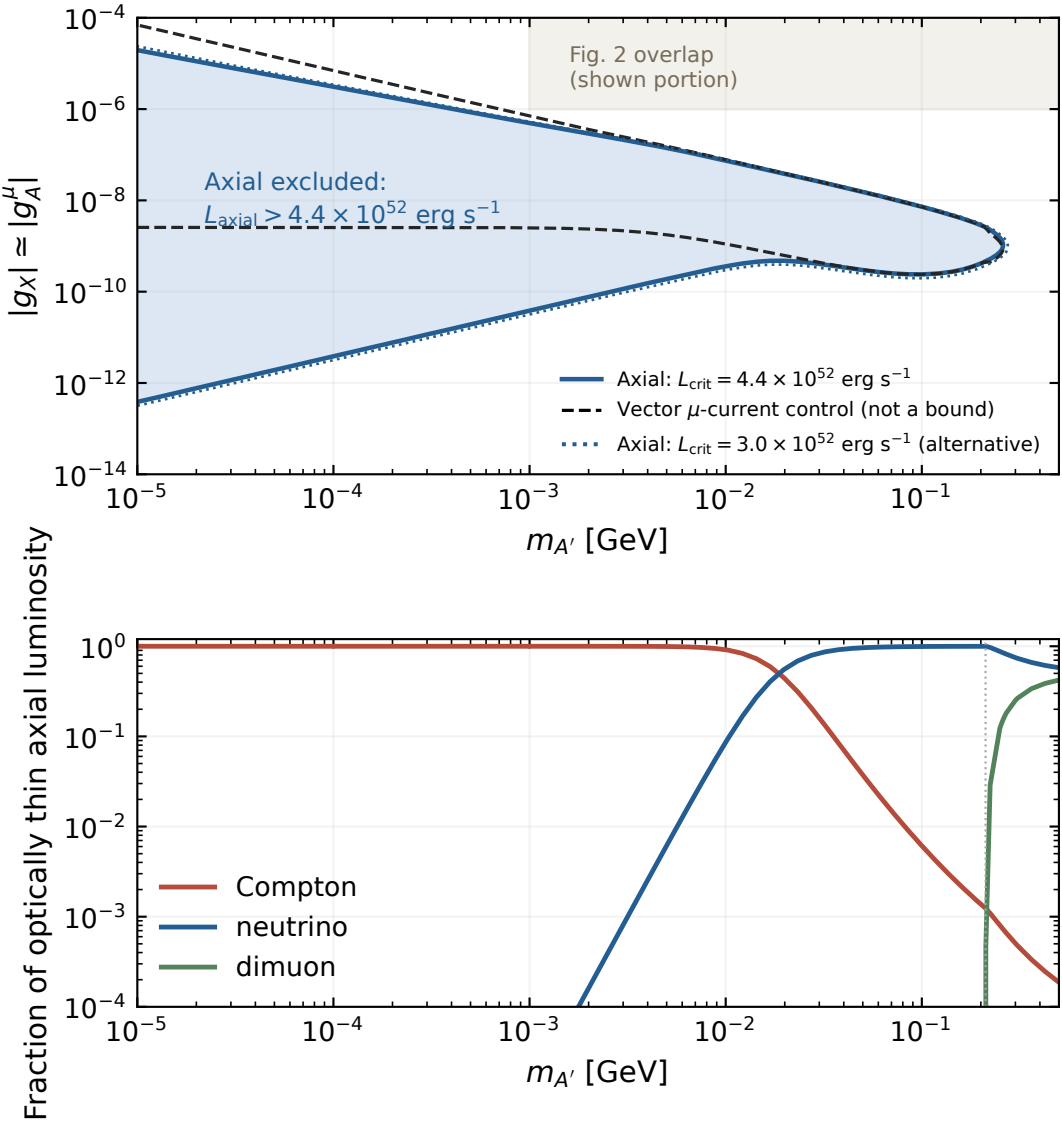
SN1987A: axial-EFT fixed-profile benchmark (GeV)
Fig. 2 overlap
(shown portion)
Axial excluded:
$L_{\rm axial} > 4.4 \times 10^{52}$ erg s^{-1}
Axial: $L_{\rm crit} = 4.4 \times 10^{52}$ erg s^{-1}
Vector μ-current control (not a bound)
Axial: $L_{\rm crit} = 3.0 \times 10^{52}$ erg s^{-1} (alternative)
$|g_\chi| \simeq |g_A^\mu|$
$m_{A'}$ [GeV]
Fraction of optically thin axial luminosity
Compton
neutrino
dimuon
$m_{A'}$ [GeV]